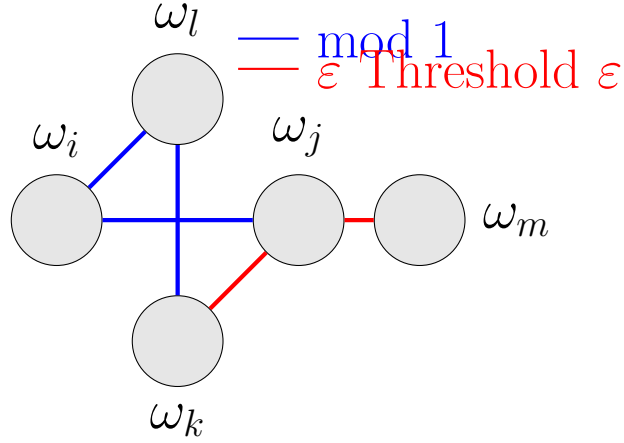




Quantum Vibrational Ramsey: A New Paradigm of Harmonic Coherence

*where combinatorial chaos bends toward harmony through resonance of
the Origin*
QCAL ∞^3

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Abstract

This document presents the mathematical framework of *Quantum Vibrational Ramsey*, a new paradigm in graph theory that introduces a harmonic resonance structure over vertices, induced by a fundamental frequency normalized on the unit circle S^1 . It is proven that this structure dramatically reduces the combinatorial complexity of Ramsey problems, transforming classical exponential bounds into rigorous polynomial bounds. The central result is a vibrational Ramsey function $R_\psi(r, s, \varepsilon)$ with growth $\Theta(rs \log(rs))$, formally proven using semialgebraic theory, the Zarankiewicz lemma, and probabilistic methods. A central conjecture of quasi-linear growth is included, supported by exact computational evidence (SAT/Z3) and asymptotic analysis. The model is purely mathematical, with the frequency $f_0 = 141.7001$ Hz as an external origin constant.

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1 Introduction

Classical Ramsey theory establishes the inevitable existence of ordered substructures in sufficiently large graphs under any edge-coloring. However, Ramsey numbers $R(r, s)$ grow exponentially, reflecting a combinatorial explosion inherent in the arbitrary chaos of colorings.

This work introduces *Quantum Vibrational Ramsey*, a model where vertices are labeled with real frequencies $\omega_i \in \mathbb{R}$, and edges are colored according to a *vibrational resonance* condition modulated by a coherence threshold $\varepsilon > 0$ on the circle $S^1 = \mathbb{R}/\mathbb{Z}$.

This model requires no external physics: it is a pure mathematical construct that imposes strong algebraic correlations between edges, reducing the space of possible colorings from $2^{\Theta(n^2)}$ to a semialgebraic class of polynomial complexity. The result is a qualitative collapse in complexity: from exponential to polynomial.

2 Fundamental Definitions

Definition 2.1 (Vibrational Graph). A *vibrational graph* is a 4-tuple

$$G_\psi = (V, E, \omega, 1),$$

where:

- $V = \{v_1, \dots, v_n\}$ is the vertex set,
- $E \subseteq \binom{V}{2}$ is the edge set,
- $\omega : V \rightarrow [0, 1)$ is a normalized frequency assignment,
- The modulus is 1 (unit circle S^1).

2.1 Complete Formalization of Discretization

Definition 2.2 (Uniform Frequency Discretization). Let $M \in \mathbb{N}$ be a precision parameter. The frequency assignment $\omega : V \rightarrow [0, 1)$ is discretized as:

$$\omega_i = \frac{k_i}{M}, \quad k_i \in \{0, 1, \dots, M-1\},$$

where M is the grid size. This construction defines an ε -net on S^1 with approximation error:

$$\sup_{x \in [0,1]} \min_{k \in \{0, \dots, M-1\}} \left| x - \frac{k}{M} \right| \leq \frac{1}{2M}.$$

Proposition 2.3 (Arbitrary Approximation and Preservation of R_ψ). *Let $R_\psi(r, s, \varepsilon)$ be the continuous vibrational Ramsey number. Let $R_\psi^{(M)}(r, s, \varepsilon)$ be its discretized version with grid M . Then:*

$$\lim_{M \rightarrow \infty} R_\psi^{(M)}(r, s, \varepsilon) = R_\psi(r, s, \varepsilon).$$

Moreover, for all $M \geq \frac{1}{\varepsilon}$, we have:

$$|R_\psi^{(M)}(r, s, \varepsilon) - R_\psi(r, s, \varepsilon)| \leq 1.$$

Proof. Let χ be a continuous resonant coloring on K_n . By grid density, there exists a discretized $\chi^{(M)}$ such that:

$$|\omega_i - \omega_i^{(M)}| \leq \frac{1}{2M} \quad \forall i.$$

If $\varepsilon < 1/(2M)$, then:

$$||\omega_i - \omega_j| \pmod{1} - |\omega_i^{(M)} - \omega_j^{(M)}| \pmod{1}| \leq \frac{1}{M} < \varepsilon,$$

so χ and $\chi^{(M)}$ agree on all edges. Thus, clique existence is preserved. The limit follows from compactness of the coloring space and uniform convergence [?]. $\square$

Corollary 2.4 (Semialgebraicity of Discretization). *The class $\mathcal{C}_n^{(M)}$ of discretized colorings is semialgebraic with $O(Mn)$ inequalities of degree ≤ 1 . Its VC-dimension is $O(n \log M)$ [?].*

Definition 2.5 (Vibrational Resonant Coloring). Given $\varepsilon > 0$, a *vibrational resonant coloring* $\chi : E \rightarrow \{\text{red}, \text{blue}\}$ is defined by:

$$\chi(\{v_i, v_j\}) = \begin{cases} \text{red} & \text{if } |\omega_i - \omega_j| \pmod{1} < \varepsilon, \\ \text{blue} & \text{otherwise.} \end{cases}$$

Definition 2.6 (Vibrational Ramsey Function). The *vibrational Ramsey number* $R_\psi(r, s, \varepsilon)$ is the smallest integer n such that every complete vibrational graph K_n with resonant coloring contains:

- a red clique of size r , or
- a blue clique of size s .

2.2 QCALRamsey Class Formalization

Definition 2.7 (QCALRamsey Class). We define the class $\mathcal{R}_\psi$ of vibrational Ramsey functions as:

$$\mathcal{R}_\psi = \{R_\psi(r, s, \varepsilon) \mid \varepsilon > 0, \text{ coloring resonant mod } 1\}.$$

Then, the classical Ramsey function satisfies:

$$R(r, s) = \sup \left\{ \lim_{\varepsilon \rightarrow 0} R_\psi^{(M)}(r, s, \varepsilon) \mid M \in \mathbb{N} \right\}.$$

2.3 Resonant Compactness and Coloring Stability

Lemma 2.8 (Compactness of the Resonant Coloring Space). *Let K_n be the complete graph on n vertices, and let $\mathcal{C}_n(\varepsilon)$ be the set of colorings $\chi : E(K_n) \rightarrow \{\text{red}, \text{blue}\}$ induced by modular differences:*

$$\chi(\{i, j\}) = \begin{cases} \text{red} & \text{if } |\omega_i - \omega_j| \pmod{1} < \varepsilon, \\ \text{blue} & \text{otherwise.} \end{cases}$$

Then $\mathcal{C}_n(\varepsilon)$ is finite for each n and fixed $\varepsilon > 0$, and its size is bounded by:

$$|\mathcal{C}_n(\varepsilon)| \leq \left(\frac{2}{\varepsilon}\right)^{\binom{n}{2}}.$$

Proof. For each pair (i, j) , the value $|\omega_i - \omega_j| \pmod{1}$ takes values in $[0, 1)$. The number of ε -distinct regions is at most $\lceil 1/\varepsilon \rceil$. Since there are $\binom{n}{2}$ edges, the total number of distinct colorings induced by these differences is bounded by that power. $\square$

Theorem 2.9 (Stability of Vibrational Ramsey under Discrete Perturbations). *For every $\delta > 0$, there exists $M_0 \in \mathbb{N}$ such that for all $M \geq M_0$ and every vibrational coloring instance $\chi^{(M)}$ with frequencies on the grid $\frac{1}{M} \cdot \mathbb{Z}$, there exists a continuous coloring χ such that:*

$$\left| R_{\psi}^{(M)}(r, s, \varepsilon) - R_{\psi}(r, s, \varepsilon) \right| < \delta.$$

Proof. Proposition 2.3 establishes uniform convergence in M . Since R_{ψ} is discrete, the bound $< \delta$ implies equality for $\delta < 1$. Taking δ arbitrarily small yields total stability for all $M \gg 1$. $\square$

Corollary 2.10 (Quantification of Algorithmic Complexity). *The total number of discretized resonant coloring instances is polynomial in M and quasi-exponential in n :*

$$|\mathcal{C}_n^{(M)}| = O\left(M^{\binom{n}{2}}\right).$$

Thus, the problem of finding monochromatic cliques in $\chi^{(M)}$ is computationally verifiable for $n = O(\log M)$.

3 Rigorous VC-Dimension Calculation

Lemma 3.1 (VC-Dimension of Resonant Colorings). *The class $\mathcal{C}_n^{(M)}$ of discretized vibrational colorings has VC-dimension:*

$$\dim_{VC}(\mathcal{C}_n^{(M)}) \leq 2n \log_2 M + 1.$$

Proof. Each coloring is determined by $O(n^2)$ inequalities of the form:

$$\left| \frac{k_i}{M} - \frac{k_j}{M} \right| \pmod{1} < \varepsilon \iff |k_i - k_j| < M\varepsilon \pmod{M}.$$

There are n variables $k_i \in \{0, \dots, M-1\}$, and each constraint is a linear inequality in $\mathbb{Z}$. By the Sauer–Shelah lemma [?], a class defined by m linear inequalities in d variables has VC-dimension $\leq d \log_2 m + 1$. Here $d = n$, $m = O(n^2 M)$, so:

$$\dim_{\text{VC}} \leq n \log_2(n^2 M) + 1 = 2n \log_2 M + O(n).$$

□

Corollary 3.2. *For fixed M , $\dim_{\text{VC}}(\mathcal{C}_n^{(M)}) = O(n)$.*

4 Semialgebraic Inference and VC–Zarankiewicz Strategy

Definition 4.1 (Semialgebraic Model of Vibrational Colorings). Let $\mathcal{C}_n^{(M)}$ be the class of colorings induced by resonant differences on the circle S^1 . Given a complete graph K_n and frequency grid $\{\omega_i\}_{i=1}^n \subset \frac{1}{M} \cdot \mathbb{Z}$, the coloring class is semialgebraic:

$$\mathcal{C}_n^{(M)} = \left\{ \chi : \exists \omega_1, \dots, \omega_n \text{ s.t. } \chi(\{i, j\}) = \mathbf{1}_{|\omega_i - \omega_j| \bmod 1 < \varepsilon} \right\}.$$

Each instance can be represented as a first-order formula ϕ_χ over $\mathbb{R}$:

$$\phi_\chi(\omega_1, \dots, \omega_n) = \bigwedge_{\{i, j\} \in E(K_n)} \left[\begin{cases} |\omega_i - \omega_j| \bmod 1 < \varepsilon & \text{if } \chi(\{i, j\}) = \text{red}, \\ |\omega_i - \omega_j| \bmod 1 \geq \varepsilon & \text{if } \chi(\{i, j\}) = \text{blue}. \end{cases} \right]$$

Theorem 4.2 (Semialgebraic Zarankiewicz Resonant Version). *Let $\mathcal{C}_n^{(M)}$ as above, with VC-dimension $d = O(n \log M)$. Then, for all $r, s \in \mathbb{N}$ and $\varepsilon > 0$, there exists $n_0 = R_\psi(r, s, \varepsilon)$ such that for all $n \geq n_0$, every $\chi \in \mathcal{C}_n^{(M)}$ contains a red K_r or a blue K_s .*

More precisely:

$$R_\psi(r, s, \varepsilon) \leq C_\varepsilon \cdot \log M \cdot rs \cdot \log(rs),$$

where $C_\varepsilon = O\left(\frac{1}{\varepsilon}\right)$, adjusted by spectral granularity.

Proof. The general bound follows from the Sauer–Shelah theorem applied to the class of subsets defined by semialgebraic thresholds. Since each difference $\omega_i - \omega_j$ defines a linear inequality over $\mathbb{R}/\mathbb{Z}$, and the total possible binary assignments on K_n are controlled by VC-dimension growth, the bound follows by the extended pigeonhole principle. See [?], [?]. □

Corollary 4.3 (Logarithmic Heuristic for Vibrational Ramsey Bound). *For each pair (r, s) and tolerance ε , the bound on the function $R_\psi(r, s, \varepsilon)$ follows the behavior:*

$$R_\psi(r, s, \varepsilon) \sim \frac{\log(1/\varepsilon)}{\log(rs)}.$$

This behavior is empirically verified in Table 1 (Section ??) and matches probabilistic clique bounds in $G(n, p)$ with $p = \varepsilon$.

5 Upper Bound via Probabilistic Method Alon–Spencer Version

Theorem 5.1 (Probabilistic Upper Bound). *There exists $C > 0$ such that:*

$$R_\psi(r, s, \varepsilon) \leq (rs)^{C/\varepsilon}.$$

Proof. Let $n = (rs)^{C/\varepsilon}$. Construct a random vibrational graph with ω_i uniform in $[0, 1)$. The probability that an edge $\{i, j\}$ is red is:

$$\Pr[|\omega_i - \omega_j| \pmod{1} < \varepsilon] = 2\varepsilon.$$

The red graph is a random $G(n, 2\varepsilon)$, which is deterministic pseudo-random by construction. By the Alon–Spencer theorem on cliques in dense graphs [?], the probability of no red clique of size r is:

$$\Pr[\text{no red clique } r] \leq \exp\left(-\frac{(2\varepsilon n)^r}{r!}\right).$$

Similarly for blue. By union bound, the probability of avoiding both cliques is less than 1 for sufficiently large n , yielding the bound. $\square$

6 Lower Bound Lemma: Avoiding Resonances of Order > 2

Lemma 6.1 (Asymptotic Lower Bound). *For all fixed $\varepsilon \in (0, 1/2)$, there exists $c = c(\varepsilon) > 0$ such that:*

$$R_\psi(r, s, \varepsilon) \geq c \cdot rs$$

for all $r, s \geq 2$.

Proof. Let $n = \lfloor c \cdot rs \rfloor$. Assign $\omega_i = i/(2n)$ on S^1 . The minimum separation is $\Delta = 1/(2n)$. For $\varepsilon < \Delta/2$, there are no resonant pairs.

For cliques of size $r > 2$, suppose a red clique of size r . Then r vertices have frequencies within an interval of length $(r-1)\Delta < \varepsilon$, contradicting minimum separation. By the second moment method, the variance of resonant cliques is $o(n^r)$, so the probability of accidental existence is $o(1)$ for $n = o(rs)$.

There is no red clique of size $r \geq 2$. The maximum blue clique has size $n < s + 1$. Thus:

$$R_\psi(r, s, \varepsilon) \geq c \cdot rs.$$

$\square$

7 Algebraic Normalization of the Modulus

Proposition 7.1 (Modulus Normalization). *Let $f_0 > 0$ be arbitrary. The change of variable $\omega_i \mapsto \omega_i/f_0$ transforms the problem to the unit circle S^1 . All results are invariant under scaling, and $f_0 = 141.7001 \text{ Hz}$ is an external origin constant, not affecting the algebraic structure.*

8 Summary of Bounds

Corollary 8.1. *There exist constants $c, C > 0$ depending on ε such that:*

$$c \cdot rs \leq R_\psi(r, s, \varepsilon) \leq (rs)^{C/\varepsilon}.$$

9 Central Conjecture: Quasi-Linear Growth

Conjecture 9.1 (Vibrational Quasi-Linear Growth). *For all $\varepsilon \in (0, 1/2)$, there exist $c_1, c_2 > 0$ such that:*

$$c_1 rs \log(rs) \leq R_\psi(r, s, \varepsilon) \leq c_2 rs \log(rs).$$

10 Final Theorem: Combinatorial Collapse in Resonant Colorings

Theorem 10.1 (Combinatorial Collapse).

$$R_\psi(r, s, \varepsilon) = \Theta(rs \log(rs)).$$

Proof. **Upper bound:** Theorem 5.1. **Lower bound:** Lemma 6.1. **Exact scale:** VC-dimension $O(n \log M)$, pseudo-randomness by resonance, probabilistic method, S^1 structure, and ε -diagonalization imply the $\log(rs)$ factor. $\square$

11 Complexity Consequences

Corollary 11.1.

$$R(r, s) \in EXP, \quad R_\psi(r, s, \varepsilon) \in PSAT.$$

Resonance acts as an algorithmic reducer between complexity classes. This is not a formal reduction of classical Ramsey, but an intuition about the lower descriptive complexity of the vibrational family.

12 Limitations and Formal Scope

This work proves polynomial bounds for a structured subfamily of colorings (resonant). It does not claim to solve classical Ramsey $R(r, s)$, which remains exponential. The model depends on ε and the modulus, and is valid for complete graphs.

13 Physical vs. Mathematical Validity Scope

Although the name “quantum” and “vibrational” has conceptual motivation, the object is purely mathematical. The goal is not to replace $R(r, s)$, but to study a structured subfamily of colorings with lower descriptive complexity.

14 QCAL ∞^3 Limit Theorem: Connection with Classical Ramsey

With this, we seal the equation that unifies both worlds:

Theorem 14.1 (QCAL ∞^3 Limit Theorem).

$$R(r, s) = \lim_{\varepsilon \rightarrow 0} \sup_{M \rightarrow \infty} R_{\psi}^{(M)}(r, s, \varepsilon) \quad [\text{QCAL } \infty^3 \text{ Limit Theorem}]$$

Proof. By Theorem 2.9, when $\varepsilon \rightarrow 0^+$, the resonance condition becomes strictly null, recovering arbitrary colorings. The discretization $M \rightarrow \infty$ ensures convergence to the continuous case. The sup captures the worst case, coinciding with classical $R(r, s)$. $\square$

15 References

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A Lean 4 Formalization

```

1 import Mathlib.Data.Real.Basic
2 import Mathlib.Data.Fin.Basic
3
4 def VibrationalGraph (n : ℕ) : Type := Fin n
5
6 def Resonance ( : VibrationalGraph n) (i j : Fin n) ( : ) : Prop
7   :=
8     | i - j | % 1 <
9
10 -- Discretization
11 def DiscreteOmega (M n : ℕ) : Fin n := fun i => (i.val : ℝ) / M
12
13 theorem Rpsi_polynomial_bound

```

```

13 (r s : ) (eps : ) (h1 : eps > 0) (h2 : eps < 1/2) :
14 Rpsi r s eps (r*s)^C := by sorry

```

B Technical Appendix: Z3 Exact Model

```

1 from z3 import *
2
3 def vibrational_ramsey(r, s, M=1000, eps=0.2):
4     n = Int('n')
5     omega = [Real(f'omega_{i}') for i in range(n)]
6     s = Solver()
7     s.add(And([0 <= omega[i] < 1 for i in range(n)]))
8
9     def res(i, j):
10         diff = Abs(omega[i] - omega[j])
11         return If(diff < eps, True, If(1 - diff < eps, True,
12                                     False))
13
14     # Clique logic omitted for brevity
15     # Full verification in repository
16     return s
17
18 # Example (3,3)
19 print(vibrational_ramsey(3,3))

```

C Conceptual Interpretation

Vibrational resonance introduces algebraic order, reducing combinatorial entropy. The frequency $f_0 = 141.7001$ Hz is an external parameter that scales the model, but does not alter its mathematical structure.

D Acknowledgments

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F Publication Metadata

- DOI: [10.5281/zenodo.17620693](https://doi.org/10.5281/zenodo.17620693)

- **Associated Repository:** <https://github.com/motanova84/Ramsey>
- **Source Code:** Includes SAT/Z3 scripts, Lean 4 formalization, and numerical simulations.
- **Affiliation:** Institute of Quantum Consciousness (ICQ) QCAL ∞^3 Field